

Math 141, S06

Quiz 5

Section:

Name: *key*

SSN: xxx-xx-

Instructions: There are totally 1 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. Given $y = f(x)$ where $f(x) = x^2 - 3x$, $x_0 = 2$ and $x_1 = 3$.

(a) Find the average rate of change of y with respect to x over the interval $[x_0, x_1]$.

$$\begin{aligned} Y_{\text{avg}} &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{[3^2 - 3 \cdot 3] - [2^2 - 3 \cdot 2]}{3 - 2} \\ &= \frac{0 - (-2)}{1} = 2 \end{aligned}$$

(b) Find the slope of the tangent line to the graph of f at x_0 .

$$\begin{aligned} m_{\text{tan}} &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(2+h)^2 - 3(2+h)] - [2^2 - 3 \cdot 2]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[4 + 4h + h^2 - 6 - 3h] - [4 - 6]}{h} \\ &= \lim_{h \rightarrow 0} \frac{h + h^2}{h} = \lim_{h \rightarrow 0} (1 + h) = 1 \end{aligned}$$

(c) Using the result in (b) to find the equation of the tangent line to the graph of f at x_0 .

$$\begin{aligned} y - f(x_0) &= m_{\text{tan}}(x - x_0) \\ y - [2^2 - 3 \cdot 2] &= 1 \cdot (x - 2) \\ y + 2 &= x - 2 \\ y &= x - 4 \end{aligned}$$